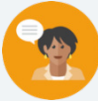


8.2 Using Properties of Parallelograms

Lesson Introduction

 Benchmarks	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.		 Learning Target	I can determine if a quadrilateral is a parallelogram using properties of a parallelogram.
	MA.912.LT.4.10 Judge the validity of arguments, and give counterexamples to disprove statements.			
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What qualities must a figure have to be a parallelogram? - How do we use properties of a parallelogram to prove a quadrilateral is a parallelogram?			
 Lesson Narrative	In this lesson, students will use the properties of parallelograms to determine if a quadrilateral is a parallelogram. Given information is provided, including in real-world scenarios, for students to examine and decide if it is enough to state the quadrilateral must be a parallelogram.			
 Component Summary	8.2.1 Warm-Up (~ 5 min.)	Complete a proof to prove if a figure is a parallelogram. (SE p. 13)	8.2.3 Guided Practice (10-15 min.)	Determine if the given information is enough to prove a figure is a parallelogram. (SE p. 16)
	8.2.2 Collaboration (10-15 min.)	Determine if quadrilaterals are parallelograms. (SE p. 14)	8.2.4 Try It (10 min.)	Use Algebra to determine if a figure is a parallelogram. (SE p. 17)
	8.2.5 Wrap-Up (~ 5 min.)	Choose properties that determine if a quadrilateral is a parallelogram. (SE p. 18)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Quadrilateral - Parallelogram		(Optional) Rubber bands (Optional) Geo Boards (Optional) Sticks (Optional) String (Optional) Rulers (Optional) Protractors	N/A

 Teacher Tips	Common Misconceptions - Students may perform incorrect calculations when substituting values for x and y in 8.2.4 Try It. - Students may set the wrong expressions equal to each other in 8.2.2 Collaboration and 8.2.4 Try It.	Things to Consider Consider using 8.2.1 Warm-Up as a formative assessment of the content understood and retained from the prior lesson.
	Literacy and Language Routines Decide and Defend For 8.2.2 Collaboration, consider utilizing a Decide and Defend routine with student partners. To utilize this routine, have students respond to each figure as to if it is a parallelogram, and have them defend their decision. Have partners (or pairs of partners) share their decisions, along with their defense statements (justifications). Allow time for students to go back and revise their answers after listening to their peers' defense.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.6.1: Reasonableness 8.2.2 Collaboration provides students the opportunity to draw conclusions about a figure and assess the reasonableness of their answer by responding to "How do you know?" through justifying.
	 Differentiate Instruction	
Lesson Notes:		

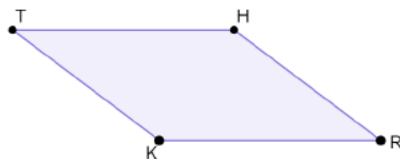
8.2.1 Warm-Up: Bell Work

Component Overview: Students will complete the proof to prove the quadrilateral with the given information is a parallelogram. Use the data gathered from this activity as a formative assessment of students' understanding of the content presented in Lesson 1.



8.2.1 Warm-Up: Bell Work

Given: $\angle RKT \cong \angle RHT$ and
 $\angle KTH \cong \angle KRH$
 Prove: $THRK$ is a
 parallelogram.



Statement	Reason
1. $\angle RKT \cong \angle RHT$ and $\angle KTH \cong \angle KRH$	1. Given
2. $m\angle RKT = m\angle RHT$ and $m\angle KTH = m\angle KRH$	2. Definition of Congruence
3. $m\angle RKT + m\angle RHT +$ $m\angle KTH + m\angle KRH = 360$	3. The sum of the angles in a quadrilateral is 360° .
4. $m\angle RKT + m\angle RKT +$ $m\angle KTH + m\angle KTH = 360^\circ$ $m\angle RHT + m\angle RHT + m\angle KRH +$ $m\angle KRH = 360^\circ$	4. <i>Substitution property of equality</i>
5. $2m\angle RKT + 2m\angle KTH = 360^\circ$ $2m\angle RHT + 2m\angle KRH = 360^\circ$	5. Addition
6. $m\angle RKT + m\angle KTH = 180^\circ$ $m\angle RHT + m\angle KRH = 180^\circ$	6. <i>Division Property of Equality</i>
7. $\angle RKT$ is supplementary to $\angle KTH$. $\angle RHT$ is supplementary to $\angle KRH$.	7. <i>Definition of Supplementary</i>
8. $\overline{TH} \parallel \overline{KR}$ and $\overline{TK} \parallel \overline{HR}$	8. If two lines are intersected by a transversal so that consecutive interior angles are supplementary, then the lines are parallel.
9. $THRK$ is a parallelogram.	9. <i>Definition of a Parallelogram</i>



Use students' understanding to inform your instruction and to determine where during this and further lessons. Make a list of misconceptions and determine which should be addressed in the whole group, and which should be addressed in a small group with those who are struggling to be proficient.

8.2.2 Collaboration: Determining Parallelograms

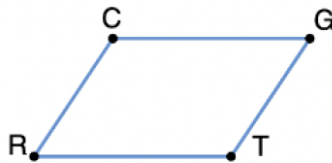
Component Overview: Students use given information to make conclusions about a series of four-sided figures. Students will justify their conclusions. Additionally, students will identify the value of two variables that will make a figure a parallelogram. Students conclude the Collaboration by summarizing the qualities a figure must have to be a parallelogram.



8.2.2 Collaboration: Determining Parallelograms

Work with your partner to determine if the figure given is a parallelogram. Finish the statements that follow.

1. $CGTR$ is shown. In the figure, $m\angle C = 113^\circ$, $m\angle G = 67^\circ$ and $m\angle T = 103^\circ$.



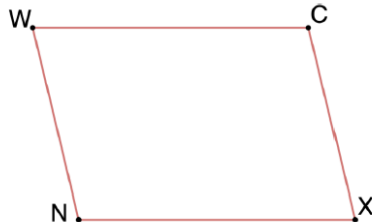
- a. I can conclude ...

that quadrilateral $CGTR$ is not a parallelogram.

- b. I know this because ...

$\angle C$ and $\angle T$ are opposite angles, but they are not congruent.

2. $WCXN$ is shown with sides $\overline{WC}$ and $\overline{NX}$ each measuring 15 cm and sides $\overline{WN}$ and $\overline{CX}$ each measuring 9 cm.



- a. I can conclude ...

that quadrilateral $WCXN$ is a parallelogram.

- b. I know this because ...

there are two pairs of opposite sides of equal length.



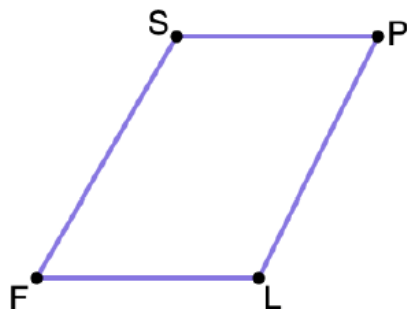
What could we change about the given information to make this figure a parallelogram?



Provide students materials to create their own quadrilateral that would fulfill the description. Supplies could include geoboards, rubber bands, string, sticks, and a ruler.

8.2.2 Collaboration: Determining Parallelograms (continued)

3. In $SPLF$, $m\angle S = 120^\circ$, and $m\angle F = (20x + 47)^\circ$. If $x = \frac{1}{2}$, determine whether the figure is a parallelogram.



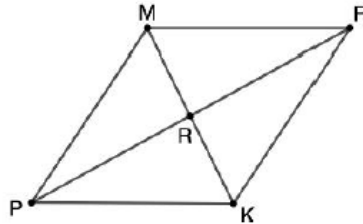
- a. I can conclude ...
 $SPLF$ is not a parallelogram.
- b. I know this because ...
 $\angle S$ and $\angle F$ are consecutive angles but are not supplementary.



After students have determined that the quadrilateral is not a parallelogram because the angles would not be supplementary, ask, "If the two angles were supplementary, would that be enough information to prove the quadrilateral is a parallelogram? (Answer: No; if $x = 0.65$, the two angles would be supplementary, but that only proves that one set of sides is parallel.)"

8.2.2 Collaboration: Determining Parallelograms (continued)

4. $MFKP$ with diagonals $\overline{MK}$ and $\overline{PF}$ intersecting at point R is shown.



Given that $MR = 4x + 7$, $KR = 3y - 2$, $FR = 4y - 3$, and $PR = 7x - 1$, find the values of x and y that make $MFKP$ a parallelogram.

$$x = \underline{\quad 6 \quad} \quad y = \underline{\quad 11 \quad}$$



Students may incorrectly set $\overline{MR} = \overline{PR}$, and $\overline{KR} = \overline{FR}$, because they share the same variable. After students create the correct equations, provide an opportunity for students to recall the methods for solving a system of equations by asking probing questions.

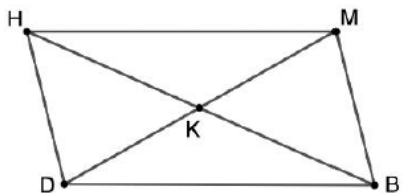
8.2.3 Guided Practice: Is It a Parallelogram?

Component Overview: Students justify whether given information is enough to determine if a quadrilateral is a parallelogram.



8.2.3 Guided Practice: Is It a Parallelogram?

- Use the figure given to fill in the chart below.



If you knew ...	...would $HMBD$ be a parallelogram?	Why or why not?
$\angle DHK \cong \angle MBK$ and $\angle BDK \cong \angle HMK$	☐ yes ☐ no	Parallel lines have congruent alternate interior angles. Two sets of parallel lines define a parallelogram.
$\overline{HM} \cong \overline{DB}$	☐ yes ☐ no	Both sets of opposite sides must be congruent, not just one set.
K is the midpoint of $\overline{DM}$ and $\overline{HB}$	☐ yes ☐ no	The midpoint is the point where two segments will bisect each other. Parallelograms have diagonals that bisect each other.
$m\angle DHK + m\angle BDK = 180^\circ$	☐ yes ☐ no	These angles are not an angle pair formed by parallel lines.



Use observational data from the 8.2.2 Collaboration to determine how much guidance is needed to address key ideas in this Guided Practice component.

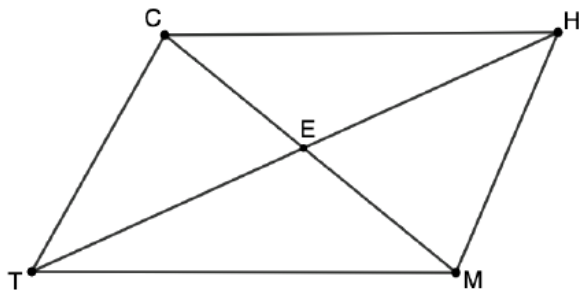


Co-Craft Questions and Problems

Consider utilizing this routine for each set of given information that had the response of “no.” Students will create assessment questions, along with the correct response, determining what additional information is needed with the given information to determine the figure is a parallelogram.

8.2.3 Guided Practice: Is It a Parallelogram? (continued)

2. Rocio recently inherited a large field in the shape of a quadrilateral from her Uncle Asher. $CHMT$ with diagonals $\overline{CM}$ and $\overline{HT}$ intersecting at point E is shown. She wants to partition the field into four triangular sections to make gardens. While she was out measuring for her supply order, she noticed that the field was shaped like a parallelogram. Her Uncle Asher disagrees. Rocio created the diagram below with the following measurements: $m\angle CHT = 23^\circ$, $m\angle CET = 72^\circ$, $m\angle CMT = 49^\circ$, and $m\angle HTM = 23^\circ$.



- a. Who is correct?

Uncle Asher

- b. How do you know?

Only one set of lines can be proven parallel from the given information. Therefore the shape is not necessarily a parallelogram.



Consider challenging students to draw examples of quadrilaterals that have the given properties. Using student drawings, guide the discussion on why the quadrilateral does not necessarily have to be a parallelogram, but could also be a non-parallelogram trapezoid.

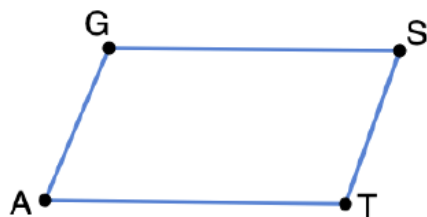
8.2.4 Try It: Parallelograms and Algebra

Component Overview: Students determine whether quadrilaterals are parallelograms using algebra to find unknown values.



8.2.4 Try It: Parallelograms and Algebra

1. Quadrilateral $GSTA$ is shown where $GS = 3x - 1$, $ST = 2y - 3$, $AT = 11$, and $GA = 8$.



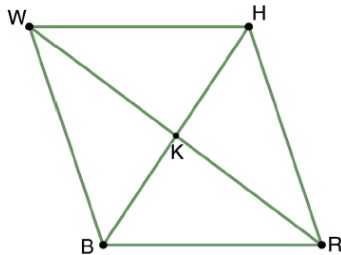
- a. If $x = 4$ and $y = 6$, determine whether $GSTA$ is a parallelogram.
No
- b. If $GSTA$ is not a parallelogram, what value(s) for x and y would make it a parallelogram?
 $x = 4$ and $y = 5.5$



Students may write the equation incorrectly by setting the wrong expressions equal to each other.

8.2.4 Try It: Parallelograms and Algebra (continued)

2. $WHRB$ is shown, where diagonals $\overline{WR}$ and $\overline{HB}$ intersect, such that $WK = -y + 4$, $RK = \frac{1}{3}x + 5$, $HK = 5x - 6$, and $BK = 11 + y$.

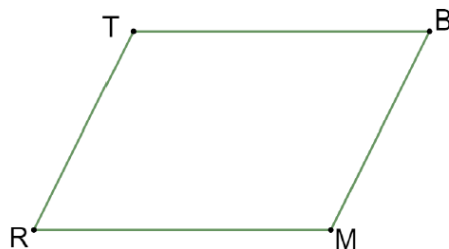


- a. If $x = 3$ and $y = -2$, determine whether $WHRB$ is a parallelogram.
Yes, it is.
- b. If $WHRB$ is not a parallelogram, what value(s) for x and y would make it a parallelogram?
No answer expected as the answer to part a is yes.



Consider allowing students to check their answers for the Try It with a partner.

3. Tariq and Agnes were having a discussion about the different properties of parallelograms. Together they drew and labeled $TBMR$, where $m\angle RTB = (5x - 2)^\circ$, $m\angle TBM = y^\circ$, and $m\angle BMR = 118^\circ$. What are the values of x and y for Tariq and Agnes' figure to be a parallelogram?



$x = 24$ and $y = 62$



If determined $y = 118$ and/or $x = 12.8$, then they used the wrong angle pair relationships, and set the wrong expressions equal to each other.

8.2.5 Wrap-Up: Cool Down

Component Overview: Students complete a checklist of properties that determine if a figure is a quadrilateral, a parallelogram, or both. Use the data collected in this activity to determine which students need remediation on the properties of parallelograms.



8.2.5 Wrap-Up: Cool Down

1. Check the property that is always true about the polygon.

Properties of Figure	Types of Polygons	
	Quadrilateral	Parallelogram
Only one set of parallel sides	☐	☐
Two sets of parallel sides	☐	☒
Two sets of sides of equal length	☐	☒
Consecutive angles are supplementary.	☐	☒
Diagonals are congruent.	☐	☐
The sum of all angles is 360° .	☒	☒
Diagonals bisect each other.	☐	☒
Opposite angles are congruent.	☐	☒



To provide an alternative entry point, have students draw as many different shapes as they can for each property.